ADETYA TINO FORRESTIAWAN

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Ponorogo Regency, East Java, Indonesia

SUMMARY

Geomatics Engineering graduate from Institut Teknologi Sepuluh Nopember focused on data engineering and analytics. Builds end-to-end data pipelines on a medallion architecture using Apache Airflow, dbt, Apache Iceberg, and Trino, including Kimball dimensional modeling with SCD Type 2 and automated data quality testing in CI. Experienced in turning heterogeneous multi-source data at the scale of hundreds of thousands of rows into BI-ready data marts, then analyzing it with Python, SQL, and Power BI into insight that supports business decisions — from credit ratio decomposition and vintage/roll rate analysis to hierarchical modeling and spatial analysis. Background in high-precision GNSS data processing on Linux at Badan Geologi (Indonesian Geological Agency).

EDUCATION

Institut Teknologi Sepuluh Nopember (ITS) — Surabaya, Indonesia
Bachelor of Engineering in Geomatics Engineering | September 2020 – February 2025
GPA: 3.06/4.00

TECHNICAL SKILLS

- **Programming & Query Languages:** Python (pandas, GeoPandas), SQL, R, JavaScript, Bash/Linux
- **Data Engineering:** Apache Airflow, dbt, Apache Iceberg, Trino, Docker, Git, GitHub Actions (CI), Soda Core
- Databases: PostgreSQL (PostGIS), DuckDB
- **Data Modeling:** Kimball dimensional modeling, Slowly Changing Dimension Type 2, medallion architecture (bronze–silver–gold), data quality testing
- **Visualization & Business Intelligence:** Microsoft Power BI, Tableau, Streamlit, Plotly, Folium, Microsoft Excel
- **Analytics & Machine Learning:** Exploratory Data Analysis, data cleaning & preprocessing, predictive analytics, Hierarchical Linear Models, spatial autocorrelation analysis (Moran's I / LISA)
- **Geospatial & Engineering:** ArcGIS, QGIS, GAMIT/GLOBK, Generic Mapping Tools, AutoCAD, Agisoft Metashape, Surpac

LICENSES & CERTIFICATIONS

- Microsoft Certified: Azure AI Fundamentals (AI-900) — Microsoft, Mar 2026
- Data Science Training — Pusdatin, Ministry of Primary and Secondary Education, 2026
- Applied Data Science — Dicoding, Apr 2026 (valid until Apr 2029)
- Junior Construction Occupational Health & Safety Expert (Level 7) — BNSP, Jun 2025 (valid until Jun 2030)
- Remote Pilot Drone — APDI/DGCA (UASTC-002), Jun 2025 (valid until Jun 2027)

PROJECTS

HUGIN — Volve Oil & Gas Field Data Lakehouse (North Sea), Independent Portfolio Project — Data Engineer | Apr 2026 – Present

- Built an end-to-end data pipeline for the open Volve field dataset (Equinor, North Sea) using Apache Airflow, dbt, Apache Iceberg, and Trino on a bronze–silver–gold medallion architecture.
- Designed a Kimball dimensional model with Slowly Changing Dimension Type 2 across 6 different source formats: production CSV, WITSML XML, LAS well logs, well directional surveys, fixed-width ASCII, and reservoir simulation reports.
- Built an identity resolution crosswalk that unifies well naming variants from 3 source systems into canonical identifiers, with coverage reporting and explicit handling of unmapped records.
- Ran automated data quality tests (dbt tests and Soda Core) in GitHub Actions on every pull request, covering grain validation, referential integrity, and domain business rules.
- Built cross-grain reconciliation controls between daily and monthly production data with automatic flagging of out-of-tolerance variances, and designed the pipeline to be idempotent and backfillable.
- Project link: https://github.com/adetyatino/hugin

BNPL Portfolio Quality & Early Warning, Independent Portfolio Project — Data Analyst | Jun 2026 – Present

- Built a Python → DuckDB → dbt analytics pipeline of 18 layered models (staging–intermediate–marts) with 152 automated data quality tests, processing a panel of 905,000 account-month rows and 527,000 transactions into 8 BI-ready data marts.
- Decomposed the headline NPL decline from 2.59% to 2.41% (May–June 2026) into rate, mix, and denominator effects reconciled against vintage curves, showing the improvement came from outstanding balance growth rather than genuine credit quality improvement.
- Produced vintage analysis, roll rates across DPD buckets, segment risk, and geographic early warning across 160 districts and cities to surface credit deterioration invisible in aggregate ratios.
- Quantified the impact of three limit tightening and approval cut-off scenarios using an Expected Loss = PD × EAD × LGD approach, presenting an explicit trade-off between avoidable losses and outstanding balance forgone.
- Designed a 5-page Power BI dashboard (Portfolio Health; Vintage; Roll Rate & Early Warning; Segment Risk; Geographic Risk); the synthetic data was calibrated against the June 2026 public release from OJK (Indonesian Financial Services Authority), with 7 of 7 metrics within ±10% tolerance and its synthetic nature openly documented in the README.
- Project link: https://github.com/adetyatino/bnpl

Literacy and Numeracy Attainment Modeling for Primary Schools in Java, Pusdatin, Ministry of Primary and Secondary Education — Data Scientist | Training Final Project | May 2026 – Aug 2026

- Mengintegrasikan dan membersihkan data lintas sumber (Rapor Pendidikan, Dapodik, BOSP, DJPK, dan IPM BPS) untuk ±65.000 SD di 119 kabupaten/kota Pulau Jawa dengan NPSN dan tabel crosswalk kode wilayah sebagai kunci penggabungan, disertai modul audit kelengkapan variabel per wilayah agar keterbatasan analisis dapat ditelusuri.
- Mengembangkan Hierarchical Linear Model dua hingga tiga level untuk memisahkan pengaruh karakteristik sekolah dan karakteristik wilayah; ICC menunjukkan 24,9% variasi literasi dan 18,7% variasi numerasi berasal dari perbedaan antarwilayah.
- Melakukan analisis autokorelasi spasial global dan lokal (Moran's I dan LISA, 999 permutasi) untuk mengidentifikasi klaster wilayah capaian rendah, antara lain di Jawa Barat bagian barat dan Madura.
- Mengembangkan dashboard interaktif 8 halaman menggunakan Python (Streamlit, GeoPandas, Folium, Plotly), mencakup ringkasan eksekutif, pemodelan, pemetaan tematik, serta unduh data dan laporan.
- Menyusun rekomendasi prioritas intervensi berdasarkan faktor paling berpengaruh dalam model: akreditasi sekolah, IPM wilayah, dan kelengkapan data pokok.
- Akses project: https://finalprojectpusdatin.streamlit.app/

WORK EXPERIENCE

Badan Geologi (Indonesian Geological Agency) — Land Surveyor and Geodynamic Data Analyst | Contract | May 2024 – Dec 2024

- Conducted GNSS surveys at 41 observation points across Surabaya and coordinated a 9-person field team for high-precision geospatial data collection.
- Processed GNSS data from 5 daily observation sessions using GAMIT/GLOBK on Linux with a script-based, repeatable processing workflow.
- Performed data quality control through Signal-to-Noise Ratio and skyplot analysis to screen out observation sessions falling below quality thresholds.
- Interpreted processing results into velocity vectors and ground deformation patterns as the basis for geodynamic monitoring of the observation area.

Institut Teknologi Sepuluh Nopember — Research Assistant | Internship | Nov 2023 – Dec 2024

- Processed survey data from the Geospatial Information Agency (BIG) using GAMIT on Linux for 16 stations over the 2014–2023 period.
- Produced a Probabilistic Seismic Hazard Analysis (PSHA) for Lampung Province based on the 2010–2021 earthquake catalog for the Sumatra source zone.
- Built automated plotting scripts with Generic Mapping Tools to generate 4 maps from processing and spatial analysis results..

ATR/BPN Surabaya 1 (National Land Agency) — Cadastral Surveyor | Internship | Aug 2023 – Sep 2023

- Conducted cadastral surveys for land title separation on 7 land parcels.
- Plotted field survey results and organized measurement archives for each certified land parcel.

LEADERSHIP ACTIVITIES

Geodesy and Geodynamics Laboratory, Kota Surabaya — Laboratory Assistant | Nov 2023 — Dec 2024

- Supported laboratory research and academic projects in geodesy and geodynamics, including data processing and team-based assignments.

Geomatics Engineering Student Association ITS, Surabaya — Event Staff, Geosentric | Oct 2021 – Aug 2024

- Served as person in charge of the 2022 ITS Geomatics Engineering graduation event.
- Produced event documentation.
- Designed the sponsorship proposal.
- Communicated association and cadre development regulations to members.

Community Service Program, Ponorogo Regency — Public Relations | Jun – Aug 2022

- Memasang PLTS dengan sistem on grid.
- Installed on-grid solar PV (PLTS) systems together with the community service team.
- Bridged the installation team and local residents, and delivered solar PV education to 34 neighborhood units (RT).